

Bayesian Econometrics

Gertraud Malsiner-Walli

WU Vienna University of Economics and Business

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Objectives

After finishing this course you

- will be familiar with the Bayesian approach to statistical learning
- know basic concepts of Bayesian statistics
- can perform a conjugate Bayesian analysis
- know how to perform Bayesian inference using Monte Carlo methods

Requirements

- Knowledge about statistical inference
- Statistical models (linear model)
- Programming in R

Today

Introduction to Bayesian inference

Statistical inference

Bayesian Approach to Statistical Inference

Conjugate Bayesian analysis

The normal model

Posterior of σ^2 with μ known

Posterior of μ with σ^2 known

Posterior of μ and σ^2

Try your luck with Monte Carlo

Introduction to Bayesian inference

Introduction to Bayesian Statistics

- Why Bayes?
- Statistical Inference
- The Bayesian Approach to Statistical Inference
- Conjugate Bayesian Analysis
 - Probability of a rare disease
 - The Normal model

Why Bayes?

Why Bayes?

Solve problems through a formula from the 18th century ...

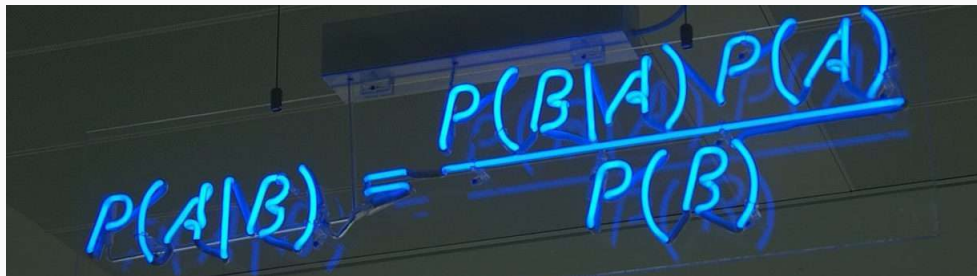


In his *Essay Towards solving a Problem in the Doctrine of Chances* (1763), **Reverend Thomas Bayes** (*1702 in London, †1761) solves the problem of inverse probabilities

→ **Bayes' Rule** and **Bayes inference** has its birth.

Timeless Bayes

... by using technology from the 21st century



A photograph of a blue neon sign mounted on a wall, displaying the Bayesian formula: $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$. The sign is illuminated with bright blue light, and the background shows a classroom environment with a chalkboard and some equipment.

International Society for Bayesian Analysis (**ISBA**):

<http://www.bayesian.org>

European Seminar of Bayesian Econometrics (**ESOBE**):

<http://www.esobe.org>

Bayes rule i

- Bayes' rule: it is based on **conditional probabilities**.
- Given two events A and B , what is $P(A)$, $P(B)$, $P(A|B)$, $P(B|A)$?
E.g. rolling a dice: $A = \text{'6'}$ and $B = \text{'number is even'}$.

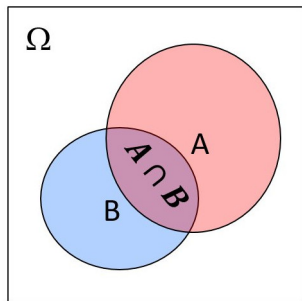


Bayes rule ii

- Bayes' rule: is a mathematical rule for **inverting conditional probabilities**, given two events A and B :

$$P(A \cap B) = P(A|B)P(B),$$

$$P(A \cap B) = P(B|A)P(A)$$



$$\Rightarrow P(B|A) = \frac{P(A \cap B)}{P(A)}$$

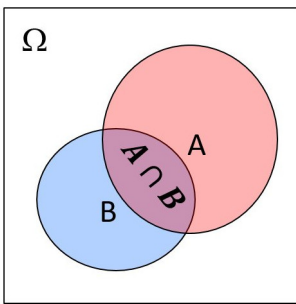
$$\Rightarrow P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

where $P(B)$ is referred to as **prior** probability and $P(B|A)$ is referred to as **posterior** probability.

Bayes rule iii

Bayes' rule if the denominator has to be calculated:

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)} = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B^c)P(B^c)}$$



Bayes rule - Example

If a bank reports large losses (L), what is the probability that the bank will default D ? $P(D|L) = ?$

- We have the following (historic) knowledge:
 - If a bank defaulted, 80% of them have reported large losses: $P(L|D) = 0.8$.
 - Prior probability that a bank defaults is $P(D) = 1/200$.
 - Prior probability of a bank reporting losses is $P(L) = 1/10$.
- If a bank reports large losses, the probability that it will default is given by:

$$P(D|L) = \frac{P(L|D)P(D)}{P(L)} = \frac{0.8 \cdot 1/200}{1/10} = 0.04$$

- People's reaction is often: "Losses? This bank is probably doomed".
- Bayes shows: "Actually, default risk rises from 0.5% to 4% - big jump, but still unlikely".

⇒ This is where Bayes' rule shines, because intuition usually overreacts to the signal and ignores how rare default is.

Exercise

- Example: Probability of Default Given a Credit Rating Downgrade
- Suppose a bank wants to assess the probability that a firm will default (A) after a credit rating downgrade (B). $P(A|B)=?$
- From historical corporate bond data (like those compiled by rating agencies such as Moody's):
 - 5% of firms default within a year. $P(A) = 0.05$
 - 70% of firms that default were downgraded beforehand. $P(B|A) = 0.7$
 - 20% of firms receive a downgrade in any given year. $P(B) = 0.2$
- What is the probability of default given a downgrade?

$$P(A|B) = (P(B|A)*P(A))/P(B) = (0.7*0.05)/0.2 = 0.175 = 17.5\%$$

Statistical inference

Statistical Inference in general

- Statistical induction:

We use a data sample to infer population characteristics

- observed data $\mathbf{y}$ quantify the outcome of a survey or experiment
- parameter θ quantifies unknown population characteristic

- Information (uncertainty)

- pre-experimentally (before seeing data) y and θ are unknown
- post-experimentally (after having seen data) $\mathbf{y}$ is known; θ is unknown

⇒ Goal: inference on the parameter θ after observing $\mathbf{y}$.

Example: Probability of a Rare Event: Corporate fraud cases

- Interest is in prevalence θ of firms committing fraud in a given year.
 - A random sample of $n = 20$ firms is checked for fraud.
 - Y is the number of firms caught committing fraud in the sample.

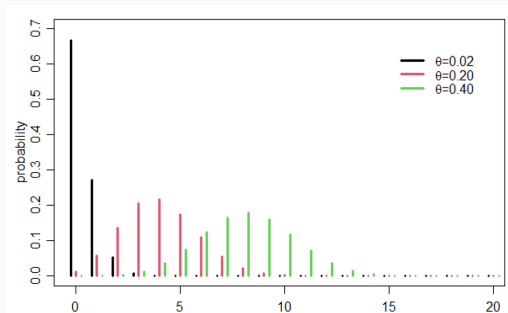
⇒ What is the information on θ ?

- Statistical model
 - Parameter space $\Theta = [0, 1]$.
 - Sample space $Y = \{0, 1, \dots, 20\}$
- For a large number of firms a reasonable sampling model for Y is

$$Y|\theta \sim \text{Binom}(20, \theta)$$

Binomial distribution

$$Y \sim \text{Binom}(n, \theta) \Rightarrow p(Y = k) = \binom{n}{k} \theta^k (1 - \theta)^{(n-k)}$$



number of caught firms committing fraud in the sample

Probability of a rare event: Statistical analysis

- Sampling model: $Y \sim \text{Binom}(20, \theta)$.
- Data: $n = 20, y = 0$.
- Statistical Tasks:
 - **Point estimation:** $\hat{\theta}$
 - **Interval estimation:** determine $[\theta_l, \theta_u]$ such that $P(\theta_l < \theta < \theta_u) = 1 - \alpha$.
 - **Hypothesis tests:** e.g. $H_0 : \theta \geq 0.1$ vs. $H_1 : \theta < 0.1$.
 - **Prediction:** number of caught $\tilde{Y}$ in a new sample of size $\tilde{n}$.

Probability of a rare event: Statistical analysis

- Sampling model: $Y \sim \text{Binom}(20, \theta)$, i.e. $f(y) = \binom{n}{y} \theta^y (1 - \theta)^{(n-y)}$
- **Point estimation**: ML- estimation
 - ML-estimate $\hat{\theta}$ is the value maximizing the probability to observe y
 - Likelihood: $P(Y = 0|\theta) = (1 - \theta)^n$
 - MLE for θ :

$$\hat{\theta} = \bar{y} = \frac{y}{n} = 0$$

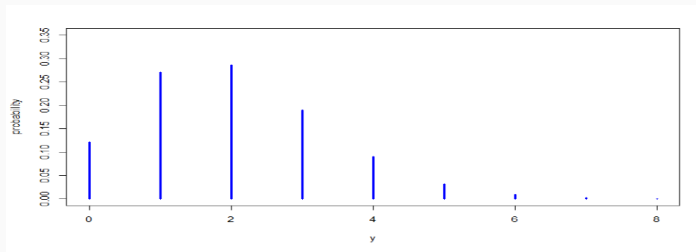
- **Interval estimation**:
 - Approximative: Wald confidence interval

$$\hat{\theta} \pm z_{1-\alpha/2} \sqrt{\hat{\theta}(1 - \hat{\theta})/n}.$$

Note: It has **length 0** if $\hat{\theta} = 0$ (for any significance level, frequentist coverage is only asymptotically correct but not for small n).

Probability of a Rare Event: Frequentist Analysis

- Hypothesis testing: $H_0 : \theta \geq 0.1$ vs. $H_1 : \theta < 0.1$.
- Test and test statistics: Distribution of Y under H_0 , i.e. for $\theta = 0.1$.



Distribution of Y under H_0 , i.e. for $\theta = 0.1$

- The p-value is the probability of observing $\mathbf{y}$ (or an even more extreme value) under H_0 :
 $\Rightarrow P(Y = 0 | \theta = 0.1) = 0.1215 \quad \Rightarrow H_0 \text{ !! (i.e. } \theta \geq 0.1)$

Probability of a Rare Event: Frequentist Analysis

- **Prediction:** conditional on $\hat{\theta}$:

$$\tilde{Y} \sim \text{Binom}(\tilde{n}, \hat{\theta})$$

Thus:

$$P(\tilde{Y} = 0 | \hat{\theta} = 0) = 1 \quad \text{for any } \tilde{n}.$$



Bayesian Approach to Statistical Inference

Bayesian approach

1. **Quantify beliefs** (information) on unknown quantities/events
⇒ beliefs can be expressed via **probability distributions**
2. **Update beliefs** in light of new information ⇒ information update (inductive learning) via Bayes rule

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

Replace:

'B': unknown parameter θ

'A': data y

$$\text{posterior } p(\theta|y) = \frac{\text{sampling dist. (Bin. in our example) = likelihood } p(y|\theta) \text{ prior dist. } p(\theta)}{\text{normalization constant (so that the posterior integrates to 1)} p(y)}$$

⇒ we get the prior updated by the data as the posterior

Bayesian analysis

- Bayesian modelling (quantification of information)
 - **prior distribution** $p(\theta)$: quantifies the belief that θ is the value of the population parameter for all possible $\theta \in \Theta$ (**before** observing the data!)
 - **Sample model** $p(y|\theta)$: quantifies the belief that y will be observed for all $\theta \in \Theta$ and $y \in Y$
- Bayesian inference is based only on **posterior distribution** $p(\theta|y)$
- Posterior distribution:
 - quantifies the belief that θ is the value of the population parameter for all $\theta \in \Theta$ **after** observing the data
 - combines information on θ from prior and data

Bayes Theorem (normalized)

$$p(\theta|\mathbf{y}) = \frac{p(\mathbf{y}|\theta)p(\theta)}{p(\mathbf{y})},$$

where $p(\sim)$ denotes densities for continuous random variables and probability function for discrete variables.

Bayes Theorem:

- describes how uncertainty on the parameter is changed by the information in the data.
- holds for discrete and continuous random variables $\mathbf{y}$, θ .

Bayes Theorem (unnormalized)

$$p(\theta|\mathbf{y}) \propto p(\mathbf{y}|\theta)p(\theta),$$

posterior $\propto$ likelihood $\times$ prior

The normalizing constant is given as

$$p(\mathbf{y}) = \sum_{\theta \in \Theta} p(\mathbf{y}|\theta)p(\theta) \quad \theta \dots \text{discrete}$$

$$p(\mathbf{y}) = \int_{\Theta} p(\mathbf{y}|\theta)p(\theta)d\theta \quad \theta \dots \text{continuous}$$

Conjugate Bayesian analysis

Probability of a rare event: Bayesian analysis

- Data: $n = 20, y = 0$
- Sampling model: $y \sim \text{Binom}(20, \theta)$

$\Rightarrow$ Likelihood: $\binom{n}{y} \theta^y (1 - \theta)^n = (1 - \theta)^n$

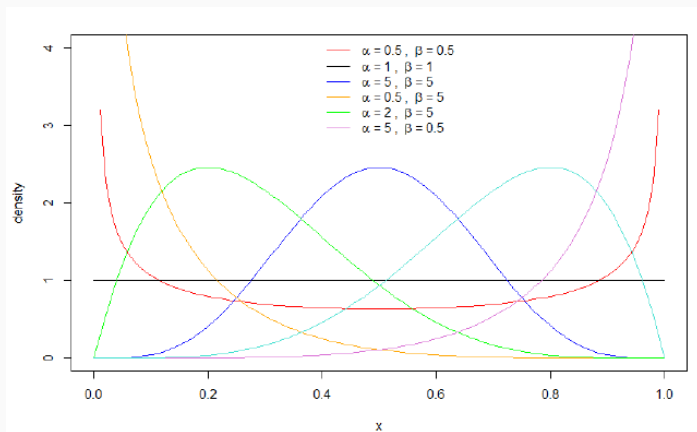
- **Of interest:** $P(\theta | Y = 0)$
- A reasonable prior distribution for $\theta \in [0, 1]$ is the Beta-distribution:

$$\theta \sim \mathcal{B}(a, b)$$

Prior: Beta distribution $\mathcal{B}(a, b)$

- Parameters: $a, b > 0$, density function:

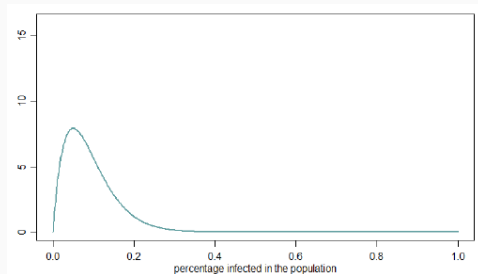
$$p(x|a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1}(1-x)^{b-1}, \quad x \in [0, 1]$$



- Moments: $E(X) = \frac{a}{a+b}$, $Var(X) = \frac{ab}{(a+b)^2(a+b+1)}$.

Prior: Selection of $\mathcal{B}(a, b)$

- Fraud rates from 0.05 to 0.2 , average prevalence of 0.1
 $\Rightarrow$ assign substantial probability to $[0.05, 0.2]$
- Specify $p(\theta) = \mathcal{B}(2, 20)$:



Note that:

$$E(\theta) = 0.09 \text{ and}$$

$$P(\theta < 0.1) = 0.64 \text{ and } P(0.05 < \theta < 0.2) = 0.66$$

Derivation of the posterior distribution

- Prior: $\theta \sim \mathcal{B}(2, 20)$
- Data: y of the n investigated firms are caught for fraud.
- The posterior $p(\theta|y)$ is derived by Bayes theorem

$$\begin{aligned} p(\theta|y) &= \frac{p(y|\theta)p(\theta)}{p(y)} \\ &= \frac{\binom{n}{y} \theta^y (1-\theta)^{n-y} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1-\theta)^{b-1} \frac{1}{p(y)}}{\theta^{a+y-1} (1-\theta)^{b+n-y-1} \cdot c(n, y, a, b)} \\ &\quad \text{constants aggregated (useful only to normalize the posterior)} \end{aligned}$$

- This is the kernel of the $\mathcal{B}(a+y, b+n-y)$ distribution:

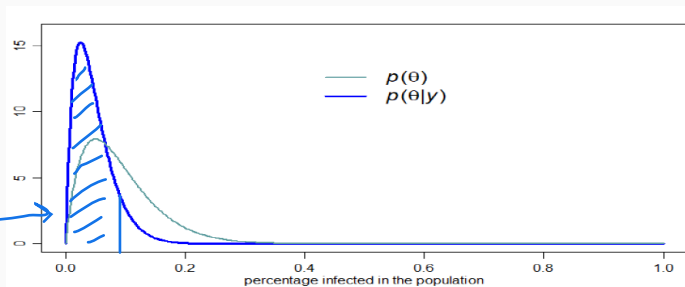
$$p(\theta|y) = \mathcal{B}(a+y, b+n-y)$$

- 'Conjugacy': **binomial** sampling combined with **Beta prior** yields a **Beta posterior**.

Posterior distribution for example

- For $n = 20$, $y = 0$, $a = 2$ and $b = 20$: we get

$$\theta|y \sim \mathcal{B}(2, 40)$$



closer to 0 than the prior,
after observing the data (0 frauds)

- Characterization of the posterior:

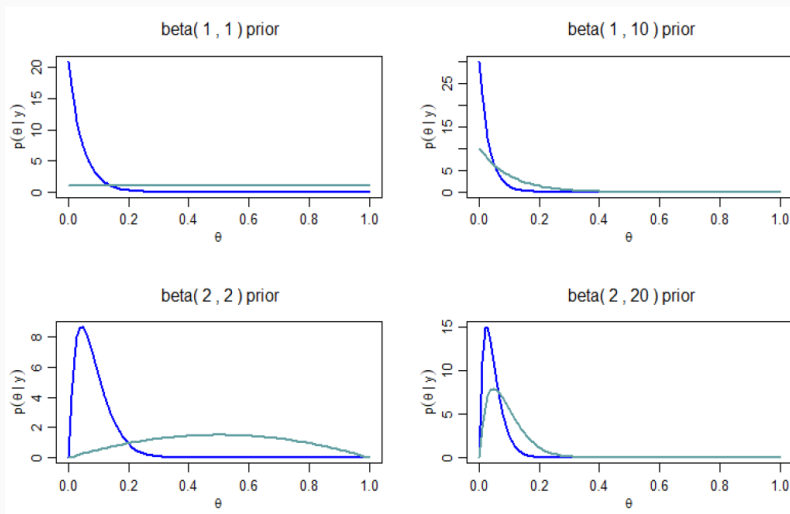
$$E(\theta|y) = 0.048$$

$$P(\theta < 0.1|y) = 0.93 \quad (\Rightarrow \text{posterior evidence for } H_1!)$$

Sensitivity Analysis

- Effect of the $\theta \sim \mathcal{B}(a, b)$ on the posterior is:

$$\theta|y \sim \mathcal{B}(a + y, b + n - y)$$



Influence of the prior

- With the posterior

$$\theta|y \sim \mathcal{B}(a + y, b + n - y)$$

the posterior expectation is

$$E(\theta|y) = \frac{a + y}{a + b + n}$$

- Without prior: $E(\theta|y) = y/n$.
- Interpretation of prior: the prior adds $a + b$ virtual observations, where a is number of successes and b the number of failures.

Point estimate

- Goal: point estimation of θ .
- Location parameter of the posterior distribution (mean, median, mode).
- Each of the three estimators is the optimal estimator with respect to a specific loss function:
 - The posterior expectation $E(\theta|\mathbf{y})$ is optimal with respect to the quadratic loss function $= (\hat{\theta} - \theta)^2$.
 - The posterior mode is optimal with respect to the 0/1 loss function $= I\{\hat{\theta} \neq \theta\}$.
 - The posterior median is optimal under the absolute deviation $= |\hat{\theta} - \theta|$.

Bayesian interval estimation

- Uncertainty on the parameter can be expressed by an **interval/region** of the posterior.
- **100 α %-credibility region**: A set C_α that contains the unknown parameter θ with a given probability α :

$$\Pr(\theta \in C_\alpha \mid \mathbf{y}) = \int_{C_\alpha} p(\theta \mid \mathbf{y}) d\theta = \alpha.$$

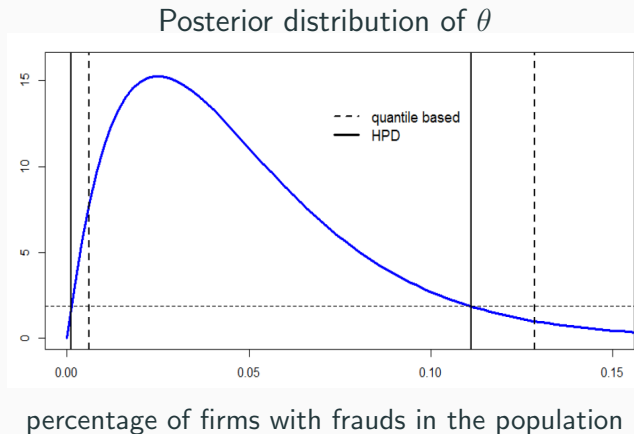
- Credibility regions:
 - Can be a **quantile based** interval:

$$[\theta_{\alpha/2}, \theta_{1-\alpha/2}]$$

- An outstanding region is **HPD region** (= highest posterior density region): it is the **smallest set among all possible regions** C_α .

Probability of a rare event: Credibility regions

- Example: 95% credibility regions for θ :
 - **Quantile based**: $[\theta_{0.025}, \theta_{0.975}] = (0.00596, 0.12855)$ (length = 0.1225)
 - **HPD region**: $(0.0011, 0.1111)$ (length = 0.1100)



Bayesian and frequentist coverage

- **Frequentist** confidence interval: $[\theta_l, \theta_u]$, where the **limits θ_l, θ_u are random variables**
 - ⇒ refers to **repeated experiments**: 95% of the confidence intervals will contain the true parameter.
 - ⇒ is **pre-experimental**: after the data are collected the confidence interval either contains the parameter or not.
- **Bayesian** credibility interval: has a **post-experimental** coverage of 95%.

⇒ there is no repeated experiments (only if we get more data → update again using Bayes rule)

Probability of a rare event: Prediction

- Goal: **prediction** of the fraud status for a firm not yet investigated.
- Let $\tilde{Y} \in 0, 1$ denote the fraud status of this firm.
- The **predictive distribution** of $\tilde{Y}$ given y :

$$\begin{aligned}P(\tilde{Y} = 1|y) &= \int_0^1 P(\tilde{Y} = 1, \theta|y) d\theta = \\&= \int_0^1 P(\tilde{Y} = 1|\theta, y) p(\theta|y) d\theta = \\&= \int_0^1 \theta p(\theta|y) d\theta = E(\theta|y) = \frac{a + y}{a + b + n} \\P(\tilde{Y} = 0|y) &= 1 - P(\tilde{Y} = 1|y) = \frac{b + n - y}{a + b + n}\end{aligned}$$

Finally: Useful distributions

Useful densities for this course:

- **Gaussian**: $p(y \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2} \left(\frac{y-\mu}{\sigma}\right)^2\right)$
- **Gamma**(shape, rate): $p(y \mid \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} \exp(-\beta y)$ support: positive numbers -> often used as the prior for modelling variance
- **Inverted Gamma**(shape, scale): $p(y \mid \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \left(\frac{1}{y}\right)^{\alpha+1} \exp\left(-\frac{\beta}{y}\right)$
- **Exponential**(rate): $p(y \mid \lambda) = \lambda \exp(-\lambda y)$
- **Generalized Student-t** (mean, scale, degrees of freedom):
$$p(y \mid \mu, \sigma, \nu) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right) \sqrt{\pi\nu}\sigma} \left(1 + \frac{1}{\nu} \left(\frac{y-\mu}{\sigma}\right)^2\right)^{-\frac{\nu+1}{2}}$$

Conjugate priors

- A prior $p(\theta)$ is **conjugate** for a sampling model $p(y_1, \dots, y_n | \theta)$ if the posterior

$$p(\theta | y_1, \dots, y_n) \propto p(y_1, \dots, y_n | \theta) p(\theta)$$

belongs to the **same** class of distributions.

- For iid observations $y_1, \dots, y_n$ conjugate priors exist e.g. for the following families $\mathcal{F}$ of distributions:

$\mathcal{F}$	parameter	conjugate prior
Binomial Binom	p	Beta $\mathcal{B}(a_0, b_0)$
Poisson $\mathcal{P}(\lambda)$	λ	Gamma $\Gamma(a_0, b_0)$
Gamma $\mathcal{G}(\alpha, \mu)$	λ	Gamma $\Gamma(a_0, b_0)$
Normal $\mathcal{N}(\mu, \sigma^2)$	μ	Normal $\mathcal{N}(b_0, B_0)$
Normal $\mathcal{N}(\mu, \sigma^2)$	σ^2	inverse Gamma $\mathcal{G}^{-1}(c_0, C_0)$

Exercise

- Derive the posterior distribution:
for the exponential distribution $Exp(\lambda)$, if the prior is a Gamma distribution $\mathcal{G}(a, b)$ (scale a , rate b).

posterior is $G(1+a, y+b)$

The normal model

The normal model

- **Data** $\mathbf{y} = (y_1, \dots, y_N)$
- **Model**: y_i are N independent observations from the normal distribution:

$$\begin{aligned}y_i &= \mu + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2) \Leftrightarrow \\y_i &\sim \mathcal{N}(\mu, \sigma^2)\end{aligned}$$

$\Rightarrow$ The likelihood function $p(\mathbf{y} \mid \mu, \sigma^2)$ is given by:

$$p(\mathbf{y} \mid \mu, \sigma^2) = \prod_{i=1}^N p(y_i \mid \mu, \sigma^2) = \left(\frac{1}{2\pi\sigma^2} \right)^{N/2} \exp \left(-\frac{1}{\sigma^2} \sum_{i=1}^N (y_i - \mu)^2 / 2 \right).$$

- Unknown parameters: μ and σ^2

Case 1: The posterior distribution of σ^2 with μ known

- For exchange rate data, it is common to assume that μ is known (typically $\mu = 0$), whereas σ^2 is unknown.

- The posterior density of σ^2 given μ is given by Bayes' theorem:

$$p(\sigma^2 \mid \mathbf{y}, \mu) \propto p(\mathbf{y} \mid \sigma^2, \mu) p(\sigma^2 \mid \mu).$$

- The likelihood function $p(\mathbf{y} \mid \sigma^2, \mu)$ is proportional to

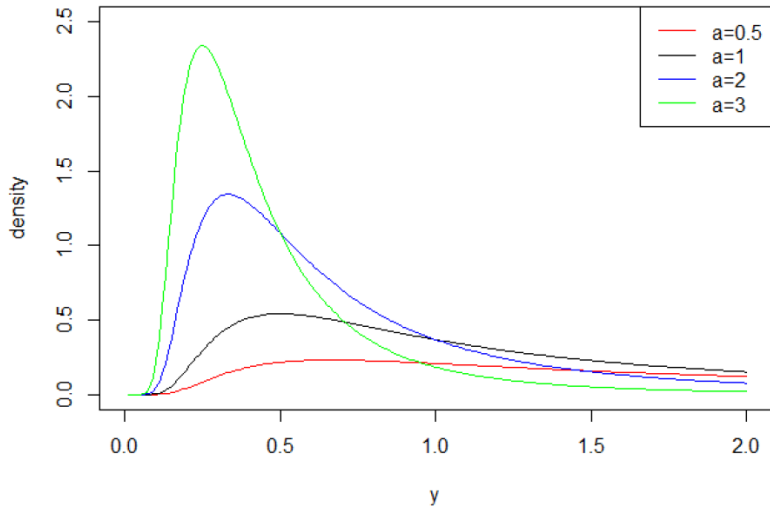
$$p(\mathbf{y} \mid \sigma^2, \mu) \propto \left(\frac{1}{\sigma^2} \right)^{N/2-1+1} \exp \left(-\frac{1}{\sigma^2} \sum_{i=1}^N (y_i - \mu)^2 / 2 \right),$$

i.e., the **pdf of an inverted Gamma distribution** (regarded as a function of σ^2).

- The pdf of inverted gamma distribution, $\sigma^2 \sim \mathcal{G}^{-1}(c, C)$, dropping constants, is:

$$p(\sigma^2; c, C) \propto \left(\frac{1}{\sigma^2} \right)^{c+1} \exp \left(-\frac{C}{\sigma^2} \right).$$

PDF of a inverse gamma distribution $\mathcal{G}^{-1}(a, b)$



Pdf of the inverse Gamma distribution (different shape a and $b=1$)

Posterior distribution of σ^2 for μ known

- As the **prior** for σ^2 choose the **inverted Gamma distribution**

$$\sigma^2 \sim \mathcal{G}^{-1}(c_0, C_0).$$

- The **posterior** distribution of σ^2 given μ is **again an inverted Gamma distribution**:

$$\sigma^2 \mid \mu, \mathbf{y} \sim \mathcal{G}^{-1}(c_N, C_N),$$

$$c_N = c_0 + N/2, \quad C_N = C_0 + \sum_{i=1}^N (y_i - \mu)^2/2.$$

- Note: Data enter through two sufficient statistics: N and the total variation of y_i around μ , $SS_T = \sum_{i=1}^N (y_i - \mu)^2$.

Case 2: The posterior distribution of μ with σ^2 known i

- **Aim:** The **posterior** density $p(\mu \mid \sigma^2, \mathbf{y})$ of μ , given σ^2 and the data $\mathbf{y}$, is:

$$p(\mu \mid \mathbf{y}, \sigma^2) \propto p(\mathbf{y} \mid \mu, \sigma^2)p(\mu \mid \sigma^2).$$

- **Likelihood:** Regarded as a function of the unknown μ , the **likelihood** $p(\mathbf{y} \mid \mu, \sigma^2)$ is (dropping all constants not depending on μ) proportional to:

$$p(\mathbf{y} \mid \mu, \sigma^2) \propto \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^N (y_i - \mu)^2 \right\}.$$

$\Rightarrow$ We use **variance decomposition**:

$$\sum_{i=1}^N (y_i - \mu)^2 = \underbrace{N(\mu - \bar{y})^2 + (N-1)s_{y,N-1}^2}_{\text{variance decomposition}}$$


where $\bar{y}$ is the mean and $s_{y,N-1}^2 = \frac{1}{N-1} \sum_{i=1}^N (y_i - \bar{y})^2$ is the (bias corrected) variance of the data.

Case 2: The posterior distribution of μ with σ^2 known ii

⇒ Regarded as a function of μ , the **likelihood** is proportional to

$$p(\mathbf{y} \mid \mu, \sigma^2) \propto \exp \left(-\frac{(\mu - \bar{y})^2}{2\sigma^2/N} \right),$$

i.e. **the pdf of a univariate normal distribution.**

- **Prior:** $\mu \sim \mathcal{N}(b_0, B_0)$

- **Posterior:**
$$\begin{aligned} p(\mu \mid \mathbf{y}, \sigma^2) &\propto \exp \left(-\frac{(\mu - \bar{y})^2}{2\sigma^2/N} \right) \exp \left(-\frac{1}{2} \frac{(\mu - b_0)^2}{B_0} \right) \\ &\propto \exp \left(-\frac{1}{2} \left[\frac{1}{\sigma^2} (N\mu^2 + N\bar{y}^2 - 2N\mu\bar{y}) + \frac{1}{B_0^2} (\mu^2 + b_0^2 - 2\mu b_0) \right] \right) \\ &\propto \exp \left(-\frac{1}{2} \left[\underbrace{\mu^2 \left(\frac{N}{\sigma^2} + \frac{1}{B_0} \right)}_{B_N^{-1}} - 2\mu \left(\frac{N\bar{y}}{\sigma^2} + \frac{b_0}{B_0} \right) \right] \right) \\ &\propto \exp \left(-\frac{1}{2} B_N^{-1} \left[\mu - B_N \left(\frac{N\bar{y}}{\sigma^2} + \frac{b_0}{B_0} \right) \right]^2 \right) \end{aligned}$$

Case 2: The posterior distribution of μ with σ^2 known iii

⇒ This is the Normal distribution

$$\mu | \sigma^2, \mathbf{y} \sim \mathcal{N}(b_n, B_n) \quad \text{with}$$
$$B_n^{-1} = \frac{1}{B_0} + \frac{N}{\sigma^2} \quad b_n = B_n \left(\frac{b_0}{B_0} + \frac{N}{\sigma^2} \bar{y} \right)$$

For more details: Homework.

Case 3: Posterior Distribution for μ and σ^2 unknown

⇒ Joint prior for μ and σ^2 required!

$$p(\mu, \sigma^2 | y) = ?$$

⇒ Now we use the vague prior for μ and σ^2 for doing closed-form analysis:

- Under the common improper prior $p(\mu, \sigma^2) \propto 1/\sigma^2$, the posterior factors into the product of

- a marginal inverted Gamma density for $\sigma^2 | \mathbf{y}$

$$\sigma^2 | \mathbf{y} \sim \mathcal{G}^{-1}(c_N, C_N) \quad c_N = (N-1)/2, \quad C_N = c_N s_{y, N-1}^2.$$

- and a conditional normal density for $\mu | \sigma^2, \mathbf{y}$ with variance proportional to σ^2 :

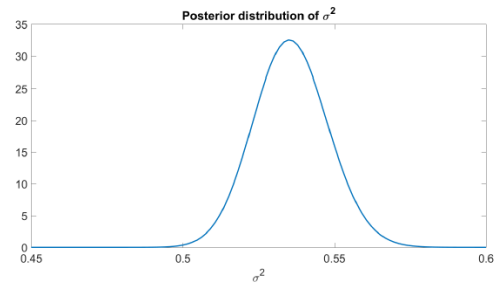
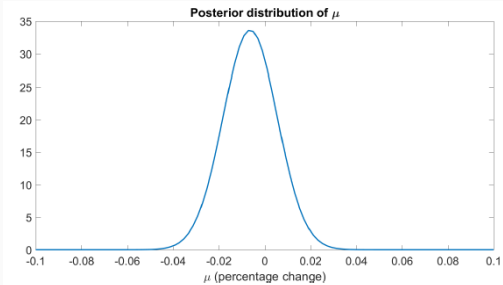
$$\mu | \sigma^2, \mathbf{y} \sim \mathcal{N}(b_N, B_N \sigma^2), \quad b_N = \bar{y}, \quad B_N = 1/N.$$

- By analytical integration we obtain $\mu | \mathbf{y} \sim t_{2c_N}(b_N, \sqrt{B_N C_N / c_N})$.
- Posterior depends on the three data summaries N , $\bar{y}$ and $s_{y, N-1}^2$.

Proof: Homework.

usually not possible to derive the posterior in a closed form solution

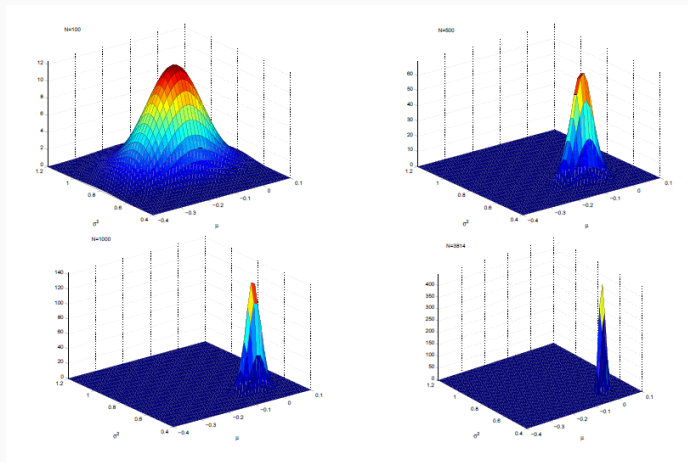
Marginal densities for SFr Exchange Rate Data



Parameter estimation

- Different location parameters of the posterior density $p(\mu, \sigma^2 | \mathbf{y})$ can be considered as point estimators for σ^2 and μ :
 - Posterior expectation of μ : $E(\mu | y) = -0.0064$, (close to 0).
 - Posterior expectation of the variance σ^2 : $E(\sigma^2) = 0.536$.
- Uncertainty quantification through second moments of the posterior:
 - Posterior variances $\text{Var}(\mu | y)$ and $\text{Var}(\sigma^2 | y)$ and standard deviations $(\sqrt{\text{Var}(\mu | y)}, \sqrt{\text{Var}(\sigma^2 | y)})$, they have similar meaning as the s.d. in ML estimation.

Bivariate posterior distribution



SFr Exchange Rate Data, modeled as i.i.d. from $\mathcal{N}(\mu, \sigma^2)$;
bivariate posterior density $p(\mu, \sigma^2 | \mathbf{y})$ based on a non-informative prior, for increasing
sample size $N = 100$, $N = 500$, $N = 1000$, and full sample size $N = 3814$.

The Posterior Distribution for μ and σ^2 unknown

What is $P(\mu > 0 \text{ and } \sigma^2 > 0.5 \mid y)$?

Try your luck with Monte Carlo

Try your luck with Monte Carlo

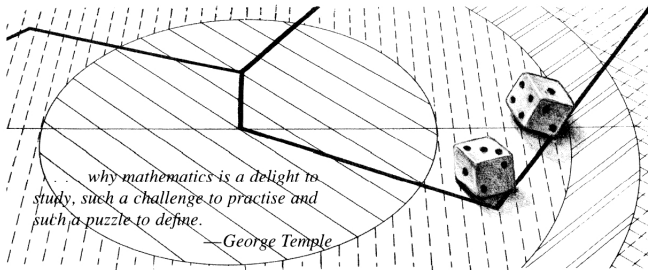
- **Bayesian computing** aims at developing efficient algorithms that allow to apply econometric methods that were impractical before the advent of computers.
- The technology behind Bayesian computing relies on:
 - **Monte Carlo simulation methods:** perform repeated random sampling from a probability distribution
 - **Laws of large numbers:** guarantees convergence of empirical moments to the moments of the probability distribution

Monte Carlo simulation methods

Developed during WW II as part of the Manhattan Project at Los Alamos by Stanislaw Ulam, John von Neumann, and Nicholas Metropolis

THE BEGINNING *of the* MONTE CARLO METHOD

by N. Metropolis



Monte Carlo Approximation of the posterior

- Idea: The posterior $p(\theta|y_1, \dots, y_N)$ can be approximated by the empirical distribution of a sample of M iid. values from the posterior:

$$\theta^{(1)}, \dots, \theta^{(M)} \sim p(\theta|y_1, \dots, y_N).$$

$\Rightarrow$ the sample $\theta^{(1)}, \dots, \theta^{(M)}$ allows to approximate the posterior of any function $g(\theta)$.

Monte Carlo Approximation

- Law of large numbers: for $M \rightarrow \infty$

$$\frac{1}{M} \sum_{m=1}^M g(\theta^{(m)}) \rightarrow E(g(\theta)|\mathbf{y}) = \int g(\theta)p(\theta|\mathbf{y})d\theta$$

- For large M therefore:

$$E(\theta|\mathbf{y}) \approx \frac{1}{M} \sum_{m=1}^M \theta^{(m)} = \bar{\theta}$$

$$\text{Var}(\theta|\mathbf{y}) \approx \frac{1}{M-1} \sum_{m=1}^M (\theta^{(m)} - \bar{\theta})^2$$

$$P(g(\theta|\mathbf{y}) \leq c) \approx \sum_{m=1}^M I(g(\theta^{(m)}) \leq c|\mathbf{y})$$

$$\theta_{\alpha} \approx \alpha\text{-quantile of } \theta^{(1)}, \dots, \theta^{(M)}$$

The Posterior Distribution for μ and σ^2 unknown

What is $P(\mu > 0 \text{ and } \sigma^2 > 0.5 \mid y)$? It's an integral, but we can appeal to the Law of Large Numbers to estimate it via Monte Carlo simulation.

Monte Carlo simulation from $p(\mu, \sigma^2 \mid y)$

Repeat the following steps for $m = 1, \dots, M$:

- (a) sample $(\sigma^2)^{(m)}$ from $p(\sigma^2 \mid y)$, which is the inverted Gamma distribution $\mathcal{G}^{-1}(c_N, C_N)$ from slide 46;
- (b) sample $\mu^{(m)}$ from $p(\mu \mid \sigma^2 = (\sigma^2)^{(m)}, y)$, which is the Gaussian distribution $\mathcal{N}(b_N, (\sigma^2)^{(m)} B_N)$ from slide 46.

To answer the question, report the proportion of indices m for which $\mu^{(m)} > 0$ and $(\sigma^2)^{(m)} > 0.5$. See `chf_usd_application1.Rmd`.

Do not forget that we are under the vague prior from slide 46. Other priors would need other algorithms.

Uncertainty quantification

- Recap: **100 α %-credibility region**: a set C_α that contains the unknown parameter θ with a given probability α :

$$\Pr(\theta \in C_\alpha \mid \mathbf{y}) = \int_{C_\alpha} p(\theta \mid \mathbf{y}) d\theta = \alpha.$$

- An outstanding region is the **highest posterior density (HPD) region**, which, for a given α , is the smallest set among all possible regions C_α .
- The 95%-H.P.D. region for μ is obtained as $[-0.0253, 0.0150]$ and covers 0. This result together with the exploratory evaluation of the posterior distribution of μ suggest that μ might be zero.
- Our rough estimation yields the 95%-H.P.D. region for σ^2 as $[0.449, 0.488]$.